

tests/chaos/ proxy_restart_recovery.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Authored + parse-clean. Detects the Squid container (READ-ONLY) and honest-SKIPs (§11.4.3) when it is absent or when the restart hook is not configured. The restart + recovery assertion runs when the **conductor** supplies the injection command.

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview

§11.4.85 / §11.4.169 **chaos** suite proving the LIVE HTTP forward proxy (Squid, localhost:53128) **recovers after a mid-flight container restart** — the process-death / §11.4.144 detect → wait → re-attach contract:

1. capture a **working** proxied request (baseline 204/200),
2. inject the fault — restart the squid container (delegated, see below),
3. poll the proxied request through a **bounded recovery window** and assert it recovers to 204/200,

emitting a **categorised recovery trace** (baseline_ok / fault_injected / probe ... state=DOWN / recovered ... state=UP) with per-event UTC timestamps. Every PASS cites the captured recovery_trace.log (§11.4.69).

Fault injection (delegated — never hardcoded)

The script **never** hardcodes podman/docker restart — the containers submodule (§11.4.76) plus the container/CI hard-stop forbid ad-hoc container commands. The restart is a **config-injected** command (§11.4.28) the conductor supplies via PROXY_CHAOS_RESTART_CMD. When it is **unset**, the fault cannot be

injected autonomously, so the suite SKIPS-with-reason (feature_disabled_by_config) — the authoring agent never restarts a container.

Prerequisites

- Committed library tests/lib/evidence.sh (sourced).
- curl, awk, grep, POSIX sh, date.
- podman OR docker for READ-ONLY container detection (optional).
- For the live assertion: a running proxy-squid container **and** PROXY_CHAOS_RESTART_CMD set to a containers-submodule restart command.
- Write access to qa-results/ (gitignored).

Usage examples

- Authoring / no-hook run (honest SKIP): `bash tests/chaos/proxy_restart_recovery.sh`
- Conductor run with the injection hook:
`PROXY_CHAOS_RESTART_CMD='<containers-submodule restart proxy-squid>' bash tests/chaos/proxy_restart_recovery.sh`
- Under host-safety caps: `GOMAXPROCS=2 nice -n 19 ionice -c 3 PROXY_CHAOS_RESTART_CMD='...' bash tests/chaos/proxy_restart_recovery.sh`

Env knobs: HTTP_PROXY_URL (default `http://localhost:53128`), HTTP_PROXY_PORT (default `53128`), CHAOS_SQUID_CONTAINER (default `proxy-squid`), CHAOS_TARGET (default `https://www.gstatic.com/generate_204`), CHAOS_EXPECT (default `204 200`), CHAOS_RECOVERY_TIMEOUT (default `60 s` — the reused reconnect budget), CHAOS_RECOVERY_POLL (default `2 s`), CURL_MAX_TIME (default `15`), PROXY_CHAOS_RESTART_CMD (injection hook), CHAOS_EVIDENCE_DIR (default `qa-results/chaos/proxy_restart_<ts>`).

Edge cases

- **Squid container not running** → SKIP (topology_unsupported). Exit 3.
- **No healthy baseline through the proxy** → SKIP (topology_unsupported) — a broken baseline is a different defect owned by the forward-proxy suites, not a recovery failure. Exit 3.
- **PROXY_CHAOS_RESTART_CMD unset** → SKIP (feature_disabled_by_config) — the fault is not injectable autonomously. Exit 3.
- **Recovers within the window** → PASS, citing `recovery_trace.log`. Exit 0.

- **Never recovers within CHAOS_RECOVERY_TIMEOUT** → FAIL. Exit 1.

Internal behaviour

- `#!/usr/bin/env bash`, POSIX-clean body (`sh -n + bash -n`, §11.4.67).
- READ-ONLY `podman ps` / `docker ps` detection (whole-name match).
- Bounded poll loop uses the reused reconnect budget (never invented, §11.4.6): poll every `CHAOS_RECOVERY_POLL` s up to `CHAOS_RECOVERY_TIMEOUT` s.
- `trace()` writes categorised timestamped events to `recovery_trace.log`.
- `trap ... EXIT INT TERM` leaves the target quiescent (§11.4.14).
- Resources: shell + curl only, single-threaded poll, well under §12.6.

Related

- `tests/lib/evidence.sh` — sourced anti-bluff evidence library.
- `tests/dynamic/suites/chaos_suite.sh` — the P10 dynamic-stack chaos suite (same delegated-injection pattern).
- `tests/stress/proxy_forward_stress.sh`, `tests/security/proxy_acl_security.sh` — sibling stress + security suites.
- Constitution §11.4.85 / §11.4.169 / §11.4.144 / §11.4.69 / §11.4.28 / §11.4.76 / §11.4.14 / §11.4.6.

Last verified

2026-07-01 — authored; `sh -n + bash -n` parse-clean; no-hook run yields an honest SKIP. The restart + recovery assertion is exercised by the conductor with `PROXY_CHAOS_RESTART_CMD` set.